UE4.27.2 — Epic source-only package loader and format

> Authority: original Epic C++ only.
>
> Revision: d94b38ae3446da52224bedd2568c078f828b4039
>
> Source root: L:/Source/Engine/UE4/UE 4.27.2
>
> **INCOMPLETE — this is a source navigation reference, not full-package format coverage or UnrealDB certification.**

Pinned source integrity

Epic file	SHA-256
LinkerLoad.cpp	`39DB1F3E1A48D829E65991259EAEFB955680CCF6165D5E31B9DB862B02560569`
PackageFileSummary.cpp	`8FF60DD29D5A0E50EB1222DA0D9C5DF71E088A1F2641E8512E6F6CE05CBA4146`
ObjectResource.cpp	`5FB996D0AAFB7DC9B2615484976A487E97E1188B42E48229FDBA024BC5BC4DD5`
ObjectVersion.h	`378C49E3AF3542C87C95D7259B81FCB77A98077D438CCCFA227B2F90870A000A`
SavePackage.cpp	`F36086C193B2496982CC283AED14849695F16D5D68BB062FE3BADD00D1093202`

SavePackage2.cpp	`FCAD38167A8612068F1300419447A9BD379253057E35F0AD2E08AF82609B4200`
EngineVersion.cpp	`BCFCE393AF77F09527EF2C7BA4EA9E6550A07E7C037265FD8DD87A4E64D81DFB`
EngineVersionBase.h	`13FA34A8F2D6D5C01AC492E998AF8D25543B7ED159296CD08BCD1EB29A6DFB4C`
ObjectMacros.h	`2BCF3EB5BA3013602FB5C02325DCFEF9B427779D3391A32D0F90D5D7A12FE20F`
Linker.h	`040C05FF670E730CDC5A4CE83A1F2985A88F7ABC1E72C369FC5FC6A75CE5E2B3`

Diagram arrows show source statement order only. Every relevant C++ conditional must be followed at its cited source line.

Version symbols in the original Epic source

Every source-defined version gate below is linked to its enum declaration. No inferred numeric values are present.

Epic version symbol	Original source declaration
VER_UE4_64BIT_EXPORTMAP_SERIALSIZES	ObjectVersion.h:621
VER_UE4_ADDED_CHUNKID_TO_ASSETDATA_AND_UPACKAGE	ObjectVersion.h:149
VER_UE4_ADDED_PACKAGE_OWNER	ObjectVersion.h:635
VER_UE4_ADDED_PACKAGE_SUMMARY_LOCALIZATION_ID	ObjectVersion.h:631
VER_UE4_ADDED_SEARCHABLE_NAMES	ObjectVersion.h:619
VER_UE4_ADD_STRING_ASSET_REFERENCES_MAP	ObjectVersion.h:364
VER_UE4_CHANGED_CHUNKID_TO_BE_AN_ARRAY_OF_CHUNKIDS	ObjectVersion.h:245
VER_UE4_COOKED_ASSETS_IN_EDITOR_SUPPORT	ObjectVersion.h:569
VER_UE4_CORRECT_LICENSEE_FLAG	ObjectVersion.h:643
VER_UE4_ENGINE_VERSION_OBJECT	ObjectVersion.h:265
VER_UE4_LOAD_FOR_EDITOR_GAME	ObjectVersion.h:326
VER_UE4_NON_OUTER_PACKAGE_IMPORT	ObjectVersion.h:639

VER_UE4_OLDEST_LOADABLE_PACKAGE	ObjectVersion.h:660
VER_UE4_PACKAGE_SUMMARY_HAS_COMPATIBLE_ENGINE_VERSION	ObjectVersion.h:487
VER_UE4_PRELOAD_DEPENDENCIES_IN_COOKED_EXPORTS	ObjectVersion.h:613
VER_UE4_SERIALIZE_TEXT_IN_PACKAGES	ObjectVersion.h:517
VER_UE4_TemplateIndex_IN_COOKED_EXPORTS	ObjectVersion.h:615
VER_UE4_WORLD_LEVEL_INFO	ObjectVersion.h:39

Package linker initialization stages

Epic function: LinkerLoad.cpp:689

Step	Source	Exact Epic statement
LOADER01	LinkerLoad.cpp:715	`Status = CreateLoader(TFunction<void()>([]() {}));`
LOADER02	LinkerLoad.cpp:724	`Status = SerializePackageFileSummary();`
LOADER03	LinkerLoad.cpp:731	`Status = SerializeNameMap();`
LOADER04	LinkerLoad.cpp:738	`Status = SerializeGatherableTextDataMap();`
LOADER05	LinkerLoad.cpp:745	`Status = SerializeImportMap();`
LOADER06	LinkerLoad.cpp:752	`Status = SerializeExportMap();`
LOADER07	LinkerLoad.cpp:760	`Status = ReconstructImportAndExportMap();`
LOADER08	LinkerLoad.cpp:768	`Status = FixupImportMap();`
LOADER09	LinkerLoad.cpp:775	`Status = PopulateInstancingContext();`
LOADER10	LinkerLoad.cpp:782	`Status = FixupExportMap();`
LOADER11	LinkerLoad.cpp:789	`Status = SerializeDependsMap();`
LOADER12	LinkerLoad.cpp:796	`Status = CreateExportHash();`
LOADER13	LinkerLoad.cpp:803	`Status = FindExistingExports();`

Step	Source	Exact Epic statement
LOADER14	LinkerLoad.cpp:809	`Status = SerializePreloadDependencies();`
LOADER15	LinkerLoad.cpp:816	`Status = FinalizeCreation(ObjectNameWithOuterToExportMap);`

Conditional statements in this original Epic source path

Source	Original C++ condition
LinkerLoad.cpp:693	`if( bHasFinishedInitialization == false )`
LinkerLoad.cpp:705	`if (GEventDrivenLoaderEnabled)`
LinkerLoad.cpp:721	`if (bCanSerializePackageFileSummary)`
LinkerLoad.cpp:728	`if( Status == LINKER_Loaded )`
LinkerLoad.cpp:735	`if( Status == LINKER_Loaded )`
LinkerLoad.cpp:742	`if( Status == LINKER_Loaded )`
LinkerLoad.cpp:749	`if( Status == LINKER_Loaded )`
LinkerLoad.cpp:755	`#if WITH_TEXT_ARCHIVE_SUPPORT`
LinkerLoad.cpp:757	`if (Status == LINKER_Loaded)`
LinkerLoad.cpp:765	`if( Status == LINKER_Loaded )`
LinkerLoad.cpp:772	`if (Status == LINKER_Loaded)`

LinkerLoad.cpp:779	`if( Status == LINKER_Loaded )`
LinkerLoad.cpp:786	`if( Status == LINKER_Loaded )`
LinkerLoad.cpp:793	`if( Status == LINKER_Loaded )`
LinkerLoad.cpp:800	`if( Status == LINKER_Loaded )`
LinkerLoad.cpp:806	`if (Status == LINKER_Loaded)`
LinkerLoad.cpp:813	`if( Status == LINKER_Loaded )`
LinkerLoad.cpp:825	`if (Status == LINKER_Failed)`
LinkerLoad.cpp:828	`#if WITH_EDITOR`
LinkerLoad.cpp:830	`if (LoadProgressScope)`

Serialized fields in source order

Source	Serialized field
—	No direct SA_VALUE sites in this function

Package summary serialization

Epic function: PackageFileSummary.cpp:48

Step	Source	Exact Epic statement
SUMMARY01	PackageFileSummary.cpp:68	`Record << SA_VALUE(TEXT("Tag"), Sum.Tag);`
SUMMARY02	PackageFileSummary.cpp:71	`if (Sum.Tag == PACKAGE_FILE_TAG || Sum.Tag == PACKAGE_FILE_TAG_SWAPPED)`
SUMMARY03	PackageFileSummary.cpp:102	`const int32 CurrentLegacyFileVersion = -7;`
SUMMARY04	PackageFileSummary.cpp:104	`Record << SA_VALUE(TEXT("LegacyFileVersion"), LegacyFileVersion);`
SUMMARY05	PackageFileSummary.cpp:124	`Record << SA_VALUE(TEXT("FileVersionUE4"), Sum.FileVersionUE4);`
SUMMARY06	PackageFileSummary.cpp:129	`Sum.CustomVersionContainer.Serialize(Record.EnterField(SA_FIELD_NAME(TEXT("CustomVersions"))), GetCustomVersionFormatForArchive(LegacyFileVersion));`
SUMMARY07	PackageFileSummary.cpp:180	`Record << SA_VALUE(TEXT("TotalHeaderSize"), Sum.TotalHeaderSize);`

Step	Source	Exact Epic statement
SUMMARY08	PackageFileSummary.cpp:182	`Record << SA_VALUE(TEXT("PackageFlags"), Sum.PackageFlags);`
SUMMARY09	PackageFileSummary.cpp:196	`Record << SA_VALUE(TEXT("NameCount"), Sum.NameCount) << SA_VALUE(TEXT("NameOffset"), Sum.NameOffset);`
SUMMARY10	PackageFileSummary.cpp:208	`Record << SA_VALUE(TEXT("ExportCount"), Sum.ExportCount) << SA_VALUE(TEXT("ExportOffset"), Sum.ExportOffset);`
SUMMARY11	PackageFileSummary.cpp:209	`Record << SA_VALUE(TEXT("ImportCount"), Sum.ImportCount) << SA_VALUE(TEXT("ImportOffset"), Sum.ImportOffset);`
SUMMARY12	PackageFileSummary.cpp:210	`Record << SA_VALUE(TEXT("DependsOffset"), Sum.DependsOffset);`
SUMMARY13	PackageFileSummary.cpp:219	`Record << SA_VALUE(TEXT("SoftPackageReferencesCount"), Sum.SoftPackageReferencesCount) << SA_VALUE(TEXT("SoftPackageReferencesOffset"), Sum.SoftPackageReferencesOffset);`

Step	Source	Exact Epic statement
SUMMARY14	PackageFileSummary.cpp:227	`Record << SA_VALUE(TEXT("ThumbnailTableOffset"), Sum.ThumbnailTableOffset);`
SUMMARY15	PackageFileSummary.cpp:258	`Record << SA_VALUE(TEXT("GenerationCount"), GenerationCount);`
SUMMARY16	PackageFileSummary.cpp:320	`Record << SA_VALUE(TEXT("CompressionFlags"), Sum.CompressionFlags);`
SUMMARY17	PackageFileSummary.cpp:329	`Record << SA_VALUE(TEXT("CompressedChunks"), CompressedChunks);`
SUMMARY18	PackageFileSummary.cpp:354	`Record << SA_VALUE(TEXT("AssetRegistryDataOffset"), Sum.AssetRegistryDataOffset);`
SUMMARY19	PackageFileSummary.cpp:355	`Record << SA_VALUE(TEXT("BulkDataStartOffset"), Sum.BulkDataStartOffset);`
SUMMARY20	PackageFileSummary.cpp:383	`Record << SA_VALUE(TEXT("PreloadDependencyCount"), Sum.PreloadDependencyCount) << SA_VALUE(TEXT("PreloadDependencyOffset"), Sum.PreloadDependencyOffset);`

Conditional statements in this original Epic source path

Source	Original C++ condition
PackageFileSummary.cpp:53	`if (BaseArchive.IsLoading())`
PackageFileSummary.cpp:66	`if (bCanStartSerializing)`
PackageFileSummary.cpp:71	`if (Sum.Tag == PACKAGE_FILE_TAG || Sum.Tag == PACKAGE_FILE_TAG_SWAPPED)`
PackageFileSummary.cpp:75	`if (Sum.Tag == PACKAGE_FILE_TAG_SWAPPED)`
PackageFileSummary.cpp:80	`if (BaseArchive.ForceByteSwapping())`
PackageFileSummary.cpp:106	`if (BaseArchive.IsLoading())`
PackageFileSummary.cpp:108	`if (LegacyFileVersion < 0) // means we have modern version numbers`
PackageFileSummary.cpp:110	`if (LegacyFileVersion < CurrentLegacyFileVersion)`
PackageFileSummary.cpp:119	`if (LegacyFileVersion != -4)`
PackageFileSummary.cpp:127	`if (LegacyFileVersion <= -2)`
PackageFileSummary.cpp:132	`if (!Sum.FileVersionUE4 && !Sum.FileVersionLicenseeUE4)`
PackageFileSummary.cpp:134	`#if WITH_EDITOR`
PackageFileSummary.cpp:135	`if (!GAllowUnversionedContentInEditor)`
PackageFileSummary.cpp:158	`if (Sum.bUnversioned)`
PackageFileSummary.cpp:184	`#if WITH_EDITOR`

PackageFileSummary.cpp:185	`if (BaseArchive.IsLoading())`
PackageFileSummary.cpp:192	`if (Sum.PackageFlags & PKG_FilterEditorOnly)`
PackageFileSummary.cpp:197	`if (!BaseArchive.IsFilterEditorOnly())`
PackageFileSummary.cpp:199	`if (BaseArchive.IsSaving() || Sum.FileVersionUE4 >= VER_UE4_ADDED_PACKAGE_SUMMARY_LOCALIZATION_ID)`
PackageFileSummary.cpp:204	`if (Sum.FileVersionUE4 >= VER_UE4_SERIALIZE_TEXT_IN_PACKAGES)`
PackageFileSummary.cpp:212	`if (BaseArchive.IsLoading() && (Sum.FileVersionUE4 < VER_UE4_OLDEST_LOADABLE_PACKAGE || Sum.FileVersionUE4 > GPackageFileUE4Version))`
PackageFileSummary.cpp:217	`if (BaseArchive.IsSaving() || Sum.FileVersionUE4 >= VER_UE4_ADD_STRING_ASSET_REFERENCES_MAP)`
PackageFileSummary.cpp:222	`if (BaseArchive.IsSaving() || Sum.FileVersionUE4 >= VER_UE4_ADDED_SEARCHABLE_NAMES)`
PackageFileSummary.cpp:233	`#if WITH_EDITORONLY_DATA`
PackageFileSummary.cpp:234	`if (!BaseArchive.IsFilterEditorOnly())`
PackageFileSummary.cpp:236	`if (BaseArchive.IsSaving() || Sum.FileVersionUE4 >= VER_UE4_ADDED_PACKAGE_OWNER)`
PackageFileSummary.cpp:249	`if (BaseArchive.IsLoading() && Sum.FileVersionUE4 >= VER_UE4_ADDED_PACKAGE_OWNER && Sum.FileVersionUE4 < VER_UE4_NON_OUTER_PACKAGE_IMPORT)`

PackageFileSummary.cpp:259	`if (BaseArchive.IsLoading() && GenerationCount > 0)`
PackageFileSummary.cpp:269	`if (BaseArchive.IsLoading() && BaseArchive.IsError())`
PackageFileSummary.cpp:275	`if (Sum.GetFileVersionUE4() >= VER_UE4_ENGINE_VERSION_OBJECT)`
PackageFileSummary.cpp:277	`if (BaseArchive.IsCooking() || (BaseArchive.IsSaving() && !FEngineVersion::Current().HasChangelist()))`
PackageFileSummary.cpp:293	`if (BaseArchive.IsLoading() && EngineChangelist != 0)`
PackageFileSummary.cpp:299	`if (Sum.GetFileVersionUE4() >= VER_UE4_PACKAGE_SUMMARY_HAS_COMPATIBLE_ENGINE_VERSION)`
PackageFileSummary.cpp:301	`if (BaseArchive.IsCooking() || (BaseArchive.IsSaving() && !FEngineVersion::Current().HasChangelist()))`
PackageFileSummary.cpp:314	`if (BaseArchive.IsLoading())`
PackageFileSummary.cpp:321	`if (!FCompression::VerifyCompressionFlagsValid(Sum.CompressionFlags))`
PackageFileSummary.cpp:331	`if (CompressedChunks.Num())`
PackageFileSummary.cpp:346	`if (LegacyFileVersion > -7)`
PackageFileSummary.cpp:357	`if (Sum.GetFileVersionUE4() >= VER_UE4_WORLD_LEVEL_INFO)`
PackageFileSummary.cpp:362	`if (Sum.GetFileVersionUE4() >= VER_UE4_CHANGED_CHUNKID_TO_BE_AN_ARRAY_OF_CHUNKIDS)`

PackageFileSummary.cpp:366	`else if (Sum.GetFileVersionUE4() >= VER_UE4_ADDED_CHUNKID_TO_ASSETDATA_AND_UPACKAGE)`
PackageFileSummary.cpp:369	`if (BaseArchive.IsLoading())`
PackageFileSummary.cpp:375	`if (ChunkID >= 0)`
PackageFileSummary.cpp:381	`if (BaseArchive.IsSaving() || Sum.FileVersionUE4 >= VER_UE4_PRELOAD_DEPENDENCIES_IN_COOKED_EXPORTS)`

Serialized fields in source order

Source	Serialized field
PackageFileSummary.cpp:68	`Tag`
PackageFileSummary.cpp:104	`LegacyFileVersion`
PackageFileSummary.cpp:122	`LegacyUE3Version`
PackageFileSummary.cpp:124	`FileVersionUE4`
PackageFileSummary.cpp:125	`FileVersionLicenseeUE4`
PackageFileSummary.cpp:161	`LegacyUE3version`
PackageFileSummary.cpp:162	`FileVersionUE4`
PackageFileSummary.cpp:163	`FileVersionLicenseeUE4`
PackageFileSummary.cpp:172	`LegacyUE3Version`

PackageFileSummary.cpp:173	`FileVersionUE4`
PackageFileSummary.cpp:174	`FileVersionLicenseeUE4`
PackageFileSummary.cpp:180	`TotalHeaderSize`
PackageFileSummary.cpp:181	`FolderName`
PackageFileSummary.cpp:182	`PackageFlags`
PackageFileSummary.cpp:196	`NameCount`
PackageFileSummary.cpp:196	`NameOffset`
PackageFileSummary.cpp:201	`LocalizationId`
PackageFileSummary.cpp:206	`GatherableTextDataCount`
PackageFileSummary.cpp:206	`GatherableTextDataOffset`
PackageFileSummary.cpp:208	`ExportCount`
PackageFileSummary.cpp:208	`ExportOffset`
PackageFileSummary.cpp:209	`ImportCount`
PackageFileSummary.cpp:209	`ImportOffset`
PackageFileSummary.cpp:210	`DependsOffset`
PackageFileSummary.cpp:219	`SoftPackageReferencesCount`
PackageFileSummary.cpp:219	`SoftPackageReferencesOffset`

PackageFileSummary.cpp:224	`SearchableNamesOffset`
PackageFileSummary.cpp:227	`ThumbnailTableOffset`
PackageFileSummary.cpp:230	`Guid`
PackageFileSummary.cpp:238	`PersistentGuid`
PackageFileSummary.cpp:252	`OwnerPersistentGuid`
PackageFileSummary.cpp:258	`GenerationCount`
PackageFileSummary.cpp:280	`SavedByEngineVersion`
PackageFileSummary.cpp:284	`SavedByEngineVersion`
PackageFileSummary.cpp:291	`EngineChangelist`
PackageFileSummary.cpp:304	`CompatibleWithEngineVersion`
PackageFileSummary.cpp:308	`CompatibleWithEngineVersion`
PackageFileSummary.cpp:320	`CompressionFlags`
PackageFileSummary.cpp:329	`CompressedChunks`
PackageFileSummary.cpp:339	`PackageSource`
PackageFileSummary.cpp:344	`AdditionalPackagesToCook`
PackageFileSummary.cpp:349	`NumTextureAllocations`
PackageFileSummary.cpp:354	`AssetRegistryDataOffset`

PackageFileSummary.cpp:355	`BulkDataStartOffset`
PackageFileSummary.cpp:359	`WorldTileInfoDataOffset`
PackageFileSummary.cpp:364	`ChunkIDs`
PackageFileSummary.cpp:372	`ChunkID`
PackageFileSummary.cpp:383	`PreloadDependencyCount`
PackageFileSummary.cpp:383	`PreloadDependencyOffset`

Serialized FObjectImport

Epic function: ObjectResource.cpp:283

Step	Source	Exact Epic statement
IMPORTS01	ObjectResource.cpp:287	`Record << SA_VALUE(TEXT("ClassPackage"), I.ClassPackage);`
IMPORTS02	ObjectResource.cpp:288	`Record << SA_VALUE(TEXT("ClassName"), I.ClassName);`
IMPORTS03	ObjectResource.cpp:289	`Record << SA_VALUE(TEXT("OuterIndex"), I.OuterIndex);`
IMPORTS04	ObjectResource.cpp:290	`Record << SA_VALUE(TEXT("ObjectName"), I.ObjectName);`
IMPORTS05	ObjectResource.cpp:294	`if (Slot.GetUnderlyingArchive().UE4Ver() >= VER_UE4_NON_OUTER_PACKAGE_IMPORT && !Slot.GetUnderlyingArchive().IsFilterEditorOnly())`
IMPORTS06	ObjectResource.cpp:296	`Record << SA_VALUE(TEXT("PackageName"), I.PackageName);`
IMPORTS07	ObjectResource.cpp:302	`I.SourceLinker = NULL;`

Conditional statements in this original Epic source path

Source	Original C++ condition
ObjectResource.cpp:293	`#if WITH_EDITORONLY_DATA`
ObjectResource.cpp:294	`if (Slot.GetUnderlyingArchive().UE4Ver() >= VER_UE4_NON_OUTER_PACKAGE_IMPORT && !Slot.GetUnderlyingArchive().IsFilterEditorOnly())`
ObjectResource.cpp:300	`if (Slot.GetUnderlyingArchive().IsLoading())`

Serialized fields in source order

Source	Serialized field
ObjectResource.cpp:287	`ClassPackage`
ObjectResource.cpp:288	`ClassName`
ObjectResource.cpp:289	`OuterIndex`
ObjectResource.cpp:290	`ObjectName`
ObjectResource.cpp:296	`PackageName`

Serialized FObjectExport

Epic function: ObjectResource.cpp:105

Step	Source	Exact Epic statement
EXPORTS01	ObjectResource.cpp:110	`Record << SA_VALUE(TEXT("ClassIndex"), E.ClassIndex);`
EXPORTS02	ObjectResource.cpp:111	`Record << SA_VALUE(TEXT("SuperIndex"), E.SuperIndex);`
EXPORTS03	ObjectResource.cpp:113	`if (BaseArchive.UE4Ver() >= VER_UE4_TemplateIndex_IN_COOKED_EXPORTS)`
EXPORTS04	ObjectResource.cpp:118	`Record << SA_VALUE(TEXT("OuterIndex"), E.OuterIndex);`
EXPORTS05	ObjectResource.cpp:119	`Record << SA_VALUE(TEXT("ObjectName"), E.ObjectName);`
EXPORTS06	ObjectResource.cpp:122	`Record << SA_VALUE(TEXT("ObjectFlags"), Save);`
EXPORTS07	ObjectResource.cpp:129	`if (BaseArchive.UE4Ver() < VER_UE4_64BIT_EXPORTMAP_SERIALSIZES)`
EXPORTS08	ObjectResource.cpp:145	`Record << SA_VALUE(TEXT("bForcedExport"), E.bForcedExport);`

Step	Source	Exact Epic statement
EXPORTS09	ObjectResource.cpp:152	`Record << SA_VALUE(TEXT("PackageFlags"), E.PackageFlags);`
EXPORTS10	ObjectResource.cpp:164	`if (BaseArchive.UE4Ver() >= VER_UE4_PRELOAD_DEPENDENCIES_IN_COOKED_EXPORTS)`

Conditional statements in this original Epic source path

Source	Original C++ condition
ObjectResource.cpp:113	`if (BaseArchive.UE4Ver() >= VER_UE4_TemplateIndex_IN_COOKED_EXPORTS)`
ObjectResource.cpp:124	`if (BaseArchive.IsLoading())`
ObjectResource.cpp:129	`if (BaseArchive.UE4Ver() < VER_UE4_64BIT_EXPORTMAP_SERIALSIZES)`
ObjectResource.cpp:154	`if (BaseArchive.UE4Ver() >= VER_UE4_LOAD_FOR_EDITOR_GAME)`
ObjectResource.cpp:159	`if (BaseArchive.UE4Ver() >= VER_UE4_COOKED_ASSETS_IN_EDITOR_SUPPORT)`
ObjectResource.cpp:164	`if (BaseArchive.UE4Ver() >= VER_UE4_PRELOAD_DEPENDENCIES_IN_COOKED_EXPORTS)`

Serialized fields in source order

Source	Serialized field
ObjectResource.cpp:110	`ClassIndex`
ObjectResource.cpp:111	`SuperIndex`
ObjectResource.cpp:115	`TemplateIndex`
ObjectResource.cpp:118	`OuterIndex`
ObjectResource.cpp:119	`ObjectName`
ObjectResource.cpp:122	`ObjectFlags`
ObjectResource.cpp:132	`SerialSize`
ObjectResource.cpp:136	`SerialOffset`
ObjectResource.cpp:141	`SerialSize`
ObjectResource.cpp:142	`SerialOffset`
ObjectResource.cpp:145	`bForcedExport`
ObjectResource.cpp:146	`bNotForClient`
ObjectResource.cpp:147	`bNotForServer`
ObjectResource.cpp:150	`PackageGuid`
ObjectResource.cpp:152	`PackageFlags`
ObjectResource.cpp:156	`bNotAlwaysLoadedForEditorGame`

ObjectResource.cpp:161	`bIsAsset`
ObjectResource.cpp:166	`FirstExportDependency`
ObjectResource.cpp:167	`SerializationBeforeSerializationDependencies`
ObjectResource.cpp:168	`CreateBeforeSerializationDependencies`
ObjectResource.cpp:169	`SerializationBeforeCreateDependencies`
ObjectResource.cpp:170	`CreateBeforeCreateDependencies`

Named table readers

Epic function: LinkerLoad.cpp:1496

Step	Source	Exact Epic statement
TABLES01	LinkerLoad.cpp:1544	`*this << NameEntry;`
TABLES02	LinkerLoad.cpp:1613	`Stream.EnterElement() << *Import;`
TABLES03	LinkerLoad.cpp:1893	`Stream.EnterElement() << *Export;`
TABLES04	LinkerLoad.cpp:2118	`Stream.EnterElement() << Depends;`
TABLES05	LinkerLoad.cpp:2134	`if (bHasSerializedPreloadDependencies || Summary.PreloadDependencyCount < 1 || Summary.PreloadDependencyOffset <= 0)`

Conditional statements in this original Epic source path

Source	Original C++ condition
LinkerLoad.cpp:1501	`if (IsTextFormat())`
LinkerLoad.cpp:1511	`if (NameMap.Num() == 0 && NameCount > 0)`
LinkerLoad.cpp:1516	`if (Summary.TotalHeaderSize > 0)`
LinkerLoad.cpp:1521	`if (bIsAsyncLoader)`

LinkerLoad.cpp:1531	`if (!bFinishedPrecaching)`
LinkerLoad.cpp:1548	`if (Idx % TimeSliceGranularity == TimeSliceGranularity - 1 &&`
LinkerLoad.cpp:1598	`if (IsTextFormat())`
LinkerLoad.cpp:1603	`if(ImportMapIndex == 0 && Summary.ImportCount > 0 )`
LinkerLoad.cpp:1878	`if (IsTextFormat())`
LinkerLoad.cpp:1883	`if(ExportMapIndex == 0 && Summary.ExportCount > 0)`
LinkerLoad.cpp:2089	`if( FPlatformProperties::RequiresCookedData()`
LinkerLoad.cpp:2096	`if ( Summary.DependsOffset == 0 )`
LinkerLoad.cpp:2103	`if (DependsMapIndex == 0 && Summary.ExportCount > 0)`
LinkerLoad.cpp:2105	`if (!IsTextFormat())`

LinkerLoad.cpp:2134	`if (bHasSerializedPreloadDependencies ||`
	`Summary.PreloadDependencyCount < 1 ||`
	`Summary.PreloadDependencyOffset <= 0)`

LinkerLoad.cpp:2139	`if (!IsTextFormat())`
LinkerLoad.cpp:2146	`if ((IsSaving() // if we are saving, we always do the ordinary serialize as a way to make sure it matches up with bulk serialization`

Serialized fields in source order

Source	Serialized field
—	No direct SA_VALUE sites in this function

Classic package writer cross-check

Epic function: SavePackage.cpp:3720

Step	Source	Exact Epic statement
WRITERS01	SavePackage.cpp:3720	`Linker->Summary.DependsOffset = Linker->Tell();`
WRITERS02	SavePackage.cpp:3723	`TArray<FPackageIndex>& Depends = Linker->DependsMap[i];`
WRITERS03	SavePackage.cpp:3724	`DependsStream.EnterElement() << Depends;`
WRITERS04	SavePackage.cpp:3743	`Linker->Summary.SoftPackageReferencesCount = Linker->SoftPackageReferenceList.Num();`
WRITERS05	SavePackage.cpp:3750	`for (FName& SoftPackageName : Linker->SoftPackageReferenceList)`
WRITERS06	SavePackage.cpp:3752	`SoftReferenceStream.EnterElement() << SoftPackageName;`

Conditional statements in this original Epic source path

Source	Original C++ condition
SavePackage.cpp:3737	`if (!(Linker->Summary.PackageFlags & PKG_FilterEditorOnly))`

SavePackage.cpp:3744	`if (!bTextFormat)`
SavePackage.cpp:3746	`#if WITH_EDITOR`

Serialized fields in source order

Source	Serialized field
—	No direct SA_VALUE sites in this function

Epic UE4.26 engine-version correction inside package summary

Epic function: PackageFileSummary.cpp:33

Step	Source	Exact Epic statement
VERSION_CORRECTION01	PackageFileSummary.cpp:33	`static void FixCorruptEngineVersion(int ObjectVersion, FEngineVersion& Version)`
VERSION_CORRECTION02	PackageFileSummary.cpp:37	`if (ObjectVersion < VER_UE4_CORRECT_LICENSEE_FLAG`
VERSION_CORRECTION03	PackageFileSummary.cpp:38	`&& Version.GetMajor() == 4`
VERSION_CORRECTION04	PackageFileSummary.cpp:39	`&& Version.GetMinor() == 26`
VERSION_CORRECTION05	PackageFileSummary.cpp:40	`&& Version.GetPatch() == 0`
VERSION_CORRECTION06	PackageFileSummary.cpp:41	`&& Version.GetChangelist() >= 12740027`
VERSION_CORRECTION07	PackageFileSummary.cpp:42	`&& Version.IsLicenseeVersion())`
VERSION_CORRECTION08	PackageFileSummary.cpp:44	`Version.Set(4, 26, 0, Version.GetChangelist(), Version.GetBranch());`

Conditional statements in this original Epic source path

Source	Original C++ condition
PackageFileSummary.cpp:37	`if (ObjectVersion < VER_UE4_CORRECT_LICENSEE_FLAG`

Serialized fields in source order

Source	Serialized field
—	No direct SA_VALUE sites in this function

FEngineVersionBase licensee-bit decoding and normalisation

Epic function: EngineVersion.cpp:20

Step	Source	Exact Epic statement
ENGINE_CHANGELIST01	EngineVersion.cpp:20	`uint32 FEngineVersionBase::GetChangelist() const`
ENGINE_CHANGELIST02	EngineVersion.cpp:23	`return Changelist & (uint32)0x7fffffffU;`
ENGINE_CHANGELIST03	EngineVersion.cpp:26	`bool FEngineVersionBase::IsLicenseeVersion() const`
ENGINE_CHANGELIST04	EngineVersion.cpp:29	`return (Changelist & (uint32)0x80000000U) != 0;`
ENGINE_CHANGELIST05	EngineVersion.cpp:90	`void FEngineVersion::Set(uint16 InMajor, uint16 InMinor, uint16 InPatch, uint32 InChangelist, const FString &InBranch)`
ENGINE_CHANGELIST06	EngineVersion.cpp:95	`Changelist = InChangelist;`

Conditional statements in this original Epic source path

Source	Original C++ condition

Serialized fields in source order

Source	Serialized field
—	No direct SA_VALUE sites in this function

Editor-only transient package flag and editor filter flag

Epic function: ObjectMacros.h:102

Step	Source	Exact Epic statement
PACKAGE_FLAGS01	ObjectMacros.h:103	`enum EPackageFlags`
PACKAGE_FLAGS02	ObjectMacros.h:106	`PKG_NewlyCreated = 0x00000001, ///< Newly created package, not saved yet. In editor only.`
PACKAGE_FLAGS03	ObjectMacros.h:136	`PKG_FilterEditorOnly = 0x80000000, ///< Package has editor-only data filtered out`

Conditional statements in this original Epic source path

Source	Original C++ condition

Serialized fields in source order

Source	Serialized field
—	No direct SA_VALUE sites in this function

FLinker::FilterExport after serialization

Epic function: Linker.h:528

Step	Source	Exact Epic statement
EXPORT_FILTER01	Linker.h:528	`bool FilterExport(const FObjectExport& Export)`
EXPORT_FILTER02	Linker.h:530	`if (Export.bExportLoadFailed || Export.bWasFiltered)`
EXPORT_FILTER03	Linker.h:535	`if (!Export.bNotAlwaysLoadedForEditorGame) // Always load, even if is editor only`
EXPORT_FILTER04	Linker.h:540	`if (FilterClientButNotServer && Export.bNotForServer) // "we are a dedicated server"`
EXPORT_FILTER05	Linker.h:544	`if (FilterServerButNotClient && Export.bNotForClient) // "we are a client only"`
EXPORT_FILTER06	Linker.h:548	`if (Export.ThisIndex.IsNull()) // Export is invalid and shouldn't be processed.`
EXPORT_FILTER07	Linker.h:552	`return false;`

Conditional statements in this original Epic source path

Source	Original C++ condition

Linker.h:530	`if (Export.bExportLoadFailed || Export.bWasFiltered)`
Linker.h:534	`#if WITH_EDITOR`
Linker.h:535	`if (!Export.bNotAlwaysLoadedForEditorGame) // Always load, even if is editor only`
Linker.h:540	`if (FilterClientButNotServer && Export.bNotForServer) // "we are a dedicated server"`
Linker.h:544	`if (FilterServerButNotClient && Export.bNotForClient) // "we are a client only"`
Linker.h:548	`if (Export.ThisIndex.IsNull()) // Export is invalid and shouldn't be processed.`

Serialized fields in source order

Source	Serialized field
—	No direct SA_VALUE sites in this function

FObjectExport constructor defaults before archive version gates

Epic function: ObjectResource.cpp:30

Step	Source	Exact Epic statement
EXPORT_DEFAULTS01	ObjectResource.cpp:43	`, bForcedExport(false)`
EXPORT_DEFAULTS02	ObjectResource.cpp:44	`, bNotForClient(false)`
EXPORT_DEFAULTS03	ObjectResource.cpp:45	`, bNotForServer(false)`
EXPORT_DEFAULTS04	ObjectResource.cpp:46	`, bNotAlwaysLoadedForEditorGame(true)`
EXPORT_DEFAULTS05	ObjectResource.cpp:47	`, bIsAsset(false)`
EXPORT_DEFAULTS06	ObjectResource.cpp:48	`, bExportLoadFailed(false)`
EXPORT_DEFAULTS07	ObjectResource.cpp:50	`, bWasFiltered(false)`
EXPORT_DEFAULTS08	ObjectResource.cpp:52	`, FirstExportDependency(-1)`
EXPORT_DEFAULTS09	ObjectResource.cpp:53	`, SerializationBeforeSerializationDependencies(0)`
EXPORT_DEFAULTS10	ObjectResource.cpp:54	`, CreateBeforeSerializationDependencies(0)`
EXPORT_DEFAULTS11	ObjectResource.cpp:55	`, SerializationBeforeCreateDependencies(0)`
EXPORT_DEFAULTS12	ObjectResource.cpp:56	`, CreateBeforeCreateDependencies(0)`

Conditional statements in this original Epic source path

Source	Original C++ condition

Serialized fields in source order

Source	Serialized field
—	No direct SA_VALUE sites in this function

Unexpanded calls / format coverage

These must be traced before any whole-file compliance claim.

- CreateLoader dispatch and archive selection
- FixupImportMap / CoreRedirects and config
- FixupExportMap and class conversions
- GetPackageLinker / provider selection
- SerializePreloadDependencies full body
- FinaliseCreation / LoadAllObjects
- Object-property serialization and per-class payloads
- Bulk/UEXP/UBULK, compressed and optional data
- Platform, target, cooked, text archive, IoStore branch separation
- SavePackage2 complete writer branches
- FLinker::FilterExport target-context inputs: WITH_EDITOR, FilterClientButNotServer, FilterServerButNotClient, bExportLoadFailed and previously filtered state